



Differential impact of multiple risks on green and conventional bond markets: Evidence from multifractal analysis

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ABSTRACT

Various major events (e.g., the 2008 economic crisis, the COVID-19 pandemic, and the Russia–Ukraine war) have introduced a surge in uncertainty across various fields, causing severe harm to financial markets. From a multifractal perspective, this study explores the correlation between economic, geopolitical, market, and energy uncertainties and green and conventional bond markets. By analyzing data from the economic policy uncertainty index, geopolitical risk index, market volatility index, crude oil volatility index, S&P green bond index, and S&P conventional bond index, spanning the period from July 5, 2016, to February 27, 2023, the study employs asymmetric multifractal detrended cross-correlation analysis and multifractal detrended cross-correlation analysis based on risk transmission to investigate the complex correlations and interactions among these variables. Results reveal that the multifractal correlations between uncertainties and bond markets are heterogeneous and related to uncertainty fields, bond market types, time series trends, and size fluctuations. Large fluctuations primarily cause multifractals between uncertainties and bond markets. Moreover, the findings indicate that bond markets must pay attention to risks from the economic and geopolitical fields and can avoid the risks from the market and energy fields. The results also confirm that green bonds are generally more risk-averse than conventional bonds.

1. Introduction

As a crucial component of the financial markets, bond markets offer investors low-risk investment and financing instruments (Lee et al., 2021). In recent years, the emerging green bond market has received considerable attention from researchers (Su et al., 2022, 2023; Wei et al., 2023; Dai et al., 2023; He and Shi, 2023). The accelerated rate of contemporary global warming due to excessive greenhouse gas emissions may cause serious harm to the Earth's resources and environment in the coming decades (Naeem et al., 2022). As a type of fixed-income security, green bonds differ from conventional bonds in that the proceeds of green bonds are used exclusively to finance projects with positive environmental benefits (Naeem et al., 2021). Green bonds are a new way to advance the world economy by developing low-carbon technologies to reduce carbon emissions and have been considered one of the most promising innovations in sustainable finance (Wang et al., 2022).

Since the beginning of the 21st century, the global political and economic landscape has undergone profound changes, and the world situation has become more complex and changeable (Gu et al., 2021). Major events such as the 2008 subprime crisis, China–US

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trade friction, the COVID-19 pandemic, and the Russia–Ukraine war have dealt heavy blows to the global economic system and intensified inter-risk transmission between financial markets (Hanif et al., 2025; Lu et al., 2025). The frequent occurrence of such sudden events has made uncertainty a key factor affecting macroeconomic change and market price fluctuation (Che et al., 2024). With increased global uncertainty, a series of potential hazards have emerged, including intensified economic fluctuation, decreased market stability, and increased investment risk (Yang et al., 2023). Against this backdrop, the relatively stable income characteristics of fixed-income assets have become an essential option for investors to pursue stability when building a diversified asset allocation (Wang et al., 2025). Fixed-income assets have a multifaceted role in investment portfolios, providing stable cash flow and reducing the volatility of the overall investment portfolio through negative correlation with risky assets. This characteristic makes fixed-income assets an essential tool for improving investment portfolios' risk–return ratio to achieve more balanced asset allocation (Ponrajah and Ning, 2023; Ziadat et al., 2023). As a typical fixed-income asset, compared with commodities and stocks, bonds usually exhibit more stable return characteristics, and bonds often yield more reliable returns in the face of various risks (Guo et al., 2023). Therefore, in-depth research on the relationship between uncertainty and bond markets is essential for understanding market dynamics and guiding investment strategies in the current macroeconomic environment. This helps investors navigate the challenges of uncertainty more effectively and provides valuable information support for asset management and risk control.

The relationship between uncertainty and bond markets has received extensive research attention (Long et al., 2022; Pham and Do, 2022; Tian et al., 2022; Tang et al., 2023; Dogan et al., 2023), but some shortcomings remains. First, previous research has predominantly considered the relationship between uncertainty and bond markets from an econometric perspective; however, the interaction between the time series of different financial and economic variables are often characterized by complexity. Multifractal theory breaks from the traditional linear research paradigm to explore the complex nonlinear characteristics of the financial system from dynamics and geometry perspectives and describe the price volatility of financial markets more comprehensively and precisely (Guo et al., 2022). Extensive studies have demonstrated that most financial time series have multifractals (Gao et al., 2022; Mensi et al., 2022b; Saâdaoui, 2023). Nonetheless, previous research on uncertainty and financial markets from a fractal perspective is limited (Fernandes et al., 2022), while uncertainty has been shown to exhibit multifractal behavior (Ye et al., 2021; Aslam et al., 2022). To further understand the complex system between different areas of uncertainty and bond markets, it is essential to explore the nature of multifractal behavior, long-term memory, and related considerations between uncertainty and bond markets. Second, many studies have only considered the impact of a single or a few uncertainties on bond markets, making it necessary to examine the impact of many different areas of uncertainty on bond markets and explore the heterogeneity of the risk transmission relationship between them.

The interaction between uncertainty and bond markets may differ (Wang et al., 2024). First, different areas of uncertainty may differ and define uncertainty in different domains, with economic policy uncertainty measuring uncertainty in the economic domain, the market volatility index measuring uncertainty in the market, geopolitical risk measuring geopolitical uncertainty, and crude oil volatility measuring uncertainty in energy volatility (Lu et al., 2025). Second, diverse events can cause changes in various uncertainty indices, with natural disasters and political risk events generally having shorter impact and the timing of economic crises having a more prolonged impact (Lin et al., 2024). As a result, differences exist in the changes caused by various uncertainty indices over time, facilitating differences in the risk transmission relationship between them and financial markets (Pham and Nguyen, 2022; Huang et al., 2016). Third, the extensive body of literature has only considered the impact of uncertainty on green bonds, which served as a diversified investment vehicle for conventional bonds during COVID-19 while ignoring conventional bonds (Naeem et al., 2021). Moreover, since the changes in uncertainty indices vary due to differing major events, can green bonds function as a hedge against conventional bonds during volatility in different areas such as the economy, politics, markets, energy, and other factors? Due to different conditions of investment direction, investment period, and review requirements between conventional and green bond markets, these bonds may have different market properties and market performance; therefore, the risk transmission relationship between uncertainty and the two different bond markets may vary.

To deeply explore the complex relationship between uncertainty and bond markets, it is essential to examine the risk transmission relationship and heterogeneity between various uncertainties and bond markets. Therefore, this study addresses the following three concerns. (1) Is multifractal behavior evident between uncertainty and bond markets, and if so, what are the characteristics of the multifractal behavior between uncertainty and bond markets? (2) To examine the differences between varying uncertainties on bond markets and effectively identify the risks of different fields of uncertainty on bond markets, aiding market stakeholders in preventing the harm of uncertainty shocks on bond markets. (3) To explore the differences in the risk transmission relationship between different types of bond markets and uncertainties, existing studies have demonstrated that green bonds are a better hedging tool for a variety of assets (Gao et al., 2021; Mensi et al., 2022a). Therefore, it is crucial to examine whether green bonds can substitute as a diversified investment tool in contrast to conventional bonds when confronting uncertainty shocks from different fields.

Based on this, this study first adopts the multifractal asymmetric detrended cross-correlation analysis (MF-ADCCA) method to explore the multifractal characteristics of uncertainties and bond markets from the fractal perspective. After confirming the multifractal correlation between uncertainties and bond markets, we then employ multifractal detrended cross-correlation analysis (MF-DCCA) based on the risk conduction method to examine the risk transmission relationship between uncertainties and bond markets.

The remaining structure of this paper is arranged as follows: Section 2 discusses the relationship between uncertainties and the bond markets. Section 3 introduces in detail the multifractal method used in the study. Section 4 presents the empirical results of this study. Section 5 provides an in-depth discussion of these empirical results. Section 6 summarizes the entire study and puts forward corresponding conclusions and policy implications.

2. The relationship between uncertainties and bond markets

Different fields of uncertainty may have varying risk transmission relationships with bond markets. Economic policy uncertainty is an essential concern that reflects the severe 2008 economic downturn at the beginning of the crisis and the subsequent slow recovery (Baker et al., 2016), which has attracted wide research attention. The adverse macroeconomic impact of economic policy uncertainty is reflected in the sharp fluctuations in macroeconomic and financial asset prices, which affected the economic cycle and recovery (Fernández-Villaverde et al., 2015; Born and Pfeifer, 2014). Some economists have argued that economic policy uncertainty is an essential driver of economic recession (Bloom, 2009; Jiang et al., 2022), recovery from its negative impact is a long-term process (Gulen and Ion, 2016), and bond markets will fluctuate under the influence of economic downturns. Increased geopolitical risk raises market uncertainty and negatively affects the economic and financial environment, reducing bond markets' performance. For the green bond market, high geopolitical risk increases oil prices, encourages the adoption of green resources, and increases incentives to invest in sustainable projects (Marques et al., 2018). Furthermore, greater geopolitical risk has prompted governments to advocate for an accelerated shift to renewable energy and higher returns on green and sustainable investment (Yang and Yang, 2021; Wang et al., 2025). Market and crude oil volatility indices represent the volatility of stock and energy markets, where a negative correlation between bonds, stocks, and oil can indicate flight to or flight from quality. In general, bonds are considered safer assets than stocks and oil; therefore, bonds are generally considered high-quality assets (Opitz and Szimayer, 2018; Baele et al., 2020; Ponrajah and Ning, 2023). Flight to quality refers to investment shifting from stocks and oil to high-quality asset bonds, while flight from quality indicates investment shifts from high-quality asset bonds to stocks and oil. Therefore, assets tend to move from the falling stock or oil market to the high-quality bond market during stock and oil downturns. In contrast, a flight from quality event is likely to occur during a rise in stock and oil markets, wherein investments are reallocated from the high-quality bond market to stock and oil markets.

The same uncertainty may have differing risk transmission relationships across green and conventional bond markets. In response to increased volatility and uncertainty in financial markets in times of crisis, investors and portfolio managers with reduced risk appetites will tend to sell various risky assets, and this tendency may increase asset correlations (Baur and McDermott, 2010). When market risk or uncertainty rises, portfolios may shift from risky assets to safe havens to enhance portfolio returns and optimize risk parameters. Therefore, under different circumstances of uncertainty risk, assets may shift to more stable assets.

3. Method

3.1. Multifractal asymmetric detrended cross-correlation analysis (MF-ADCCA)

Cao et al. (2014) introduced the asymmetric detrended fluctuation analysis (A-DFA) method into the MF-DCCA method and proposed the MF-ADCCA method to analyze the asymmetric multifractal cross-correlation between two nonstationary time series.

Assume two time series $\{x^{(i)}(t)\}$ exist, where $i = 1, 2; t = 1, \dots, M$. The MF-ADCCA method steps are as follows:

Step 1: Construct the following profile sequence of two time series:

$$y^{(i)}(j) = \sum_{k=1}^j x^{(i)}(k) - \bar{x}^{(i)} \quad (1)$$

where $\bar{x}^{(i)} = (1/M)(\sum_{k=1}^M x^{(i)}(k))$, $i = 1, 2; j = 1, \dots, M$.

Step 2: Divide the profile sequence $\{y^{(i)}(j)\}$ into M_s nonoverlapping intervals with a time scale of s , where $M_s = \text{int}(M/s)$. Since the length of the sequence M may not be an integer multiple of s , to avoid data loss at the end of the sequence, repeat the process from the end of the sequence forward to obtain $2M_s$ subintervals. The original sequence and profile sequence of each subinterval are shown below:

When $v = 1, \dots, M_s; m = 1, \dots, s$.

$$x_{v,m}^{(i)} = x^{(i)}((v-1)s + m), y_{v,m}^{(i)} = y^{(i)}((v-1)s + m) \quad (2)$$

When $v = M_s + 1, \dots, 2M_s; m = 1, \dots, s$.

$$x_{v,m}^{(i)} = x^{(i)}(M - (v - M_s)s + m), y_{v,m}^{(i)} = y^{(i)}(M - (v - M_s)s + m) \quad (3)$$

Step 3: Each subinterval sequence $(x_{v,m}^{(i)})$ and $(y_{v,m}^{(i)})$ is fitted by least squares, yielding $L_{xy}^{(i)}(m) = \alpha_{xy}^{(i)} + \beta_{xy}^{(i)}$ and $L_{yy}^{(i)}(m) = \alpha_{yy}^{(i)} + \beta_{yy}^{(i)} \bullet m$ to obtain the local trend of each subinterval. Then de-trend each subinterval and the covariance function $(F_v(s))$ of each subinterval is calculated as follows:

$$F_v(s) = \frac{1}{s} \sum_{j=1}^s \prod_{i=1}^2 |y^{(i)}[(v-1)s + j] - y_v^{(i)}(j)|, v = 1, \dots, M_s \quad (4)$$

$$F_v(s) = \frac{1}{s} \sum_{j=1}^s \prod_{i=1}^2 |y^{(i)}[M - (v - M_s)s + j] - y_v^{(i)}(j)|, v = M_s + 1, \dots, 2M_s \quad (5)$$

Step 4: According to the sign of the slope $(\beta_{xy}^{(i)})$, to classify the positive and negative trends of the subinterval series $(x_{v,m}^{(i)})$. When

$\beta_{x,y}^{(i)} > 0$, it indicates that the subinterval series $(x_{v,m}^{(i)})$ exhibits an upward trend, and when $\beta_{x,y}^{(i)} < 0$, the subinterval series $(x_{v,m}^{(i)})$ indicates a negative trend. Therefore, when dividing the positive and negative trends of a time series, the q -order average fluctuation function of two time series under different trends is calculated as follows:

Overall trend:

$$F_q(s) = \left((1 / 2M_s) \sum_{v=1}^{2M_s} [F_v(s)]^{\frac{q}{2}} \right)^{\frac{1}{q}} \quad (6)$$

Upward trend:

$$F_q^+(s) = \left((1 / M^+) \sum_{v=1}^{2M_s} ((\text{sign}(\beta_{x,y}^{(i)}) + 1) / 2) [F_v(s)]^{\frac{q}{2}} \right)^{\frac{1}{q}} \quad (7)$$

Downward trend:

$$F_q^-(s) = \left((1 / M^-) \sum_{v=1}^{2M_s} ((-\text{sign}(\beta_{x,y}^{(i)}) - 1) / 2) [F_v(s)]^{\frac{q}{2}} \right)^{\frac{1}{q}} \quad (8)$$

Where $M^+ = \sum_{v=1}^{2M_s} (\text{sign}(\beta_{x,y}^{(i)}) + 1) / 2$ and $M^- = \sum_{v=1}^{2M_s} (-\text{sign}(\beta_{x,y}^{(i)}) - 1) / 2$ denote the number of sequences with upward and downward trends in sequence $x_{v,m}^{(i)}$, respectively. If for $v = 1, \dots, 2M_s$, both have $\beta_{x,y}^{(i)} \neq 0$, then have $M^+ + M^- = 2M_s$.

When power-law cross-correlation exists, the fluctuation function satisfies the following formula (Cao et al., 2014):

$$F_q(s) \sim s^{H(q)} \quad (9)$$

$$F_q^+(s) \sim s^{H^+(q)} \quad (10)$$

$$F_q^-(s) \sim s^{H^-(q)} \quad (11)$$

Taking logarithms on both sides of Eqs. (9)–(11) gives the following:

$$\log F_q(s) = H(q) \bullet \log(s) + \log A_1 \quad (12)$$

$$\log F_q^+(s) = H^+(q) \bullet \log(s) + \log A_2 \quad (13)$$

$$\log F_q^-(s) = H^-(q) \bullet \log(s) + \log A_3 \quad (14)$$

$H(q)$ is known as the Hurst exponent and is used to measure two series that are power-law correlated over time (Zhou, 2008). When $H(q) > 0.5$, the two series exhibit persistent correlation, indicating that as one series grows, the other does as well. When $H(q) < 0.5$, then the two series exhibit inversely persistent correlation, indicating that as one series grows, the other one declines. When $H^+(q) = H^-(q)$, the cross-correlation between the two series is symmetric, and vice versa indicates an asymmetric correlation. When $q > 0$, $H(q), H^+(q), H^-(q)$ measure the multifractal scaling behavior of the two series under large fluctuations in the overall, upward, and downward trends, respectively. When $q < 0$, $H(q), H^+(q), H^-(q)$ measure the respective multifractal scaling behavior of the two series under small fluctuations.

Step 4: Obtain the holder index $\alpha(q)$ and multifractal spectrum $f(\alpha)$ employing the following Legendre transformation (Gao et al., 2022; Guo et al., 2022):

$$\begin{cases} \alpha(q) = \tau'(q) \\ f(\alpha) = q[\alpha - H(q)] + 1 \end{cases} \quad (15)$$

3.2. MF-DCCA method based on risk conduction

Cao et al. (2017) proposed the MF-DCCA method based on risk transmission to examine the risk transmission relationship between financial time series, and its main steps are similar to the MF-ADCCA method. The difference is in the construction of the profile series, assuming the existence of two time series $\{x^{(1)}(t)\}$ and $\{x^{(2)}(t)\}$, if $\{x^{(2)}(t)\}$ delay time Δt and $\{x^{(1)}(t)\}$ are not delayed, then the profile series of this method is as follows:

$$y^{(1)}(m) = \sum_{t=1}^m (x^{(1)}(t) - \bar{x}^{(1)}) \quad (18)$$

$$y^{(2)}(m) = \sum_{t=1}^m (x^{(2)}(t + \Delta t) - \bar{x}^{(2)}) \quad (19)$$

where $m = 1, \dots, M - \Delta t$, $\overline{x^{(1)}} = (1/(M - \Delta t)) \sum_{t=1}^{M-\Delta t} x^{(1)}(t)$, $\overline{x^{(2)}} = (1/(M - \Delta t)) \sum_{t=1}^{M-\Delta t} x^{(2)}(t + \Delta t)$, and $\Delta t < M$.

Notably, the above profile sequence construction ignores the autocorrelation of $\{x^{(2)}(t)\}$; therefore, [Cao and Zhou \(2019\)](#) referred to the Granger causality test and introduced noise into sequence $\{y^{(2)}(m)\}$ to weaken the effect of autocorrelation. The corrected profile sequence is as follows:

$$y^{(1)}(m) = \sum_{t=1}^m (x^{(1)}(t) - \overline{x^{(1)}(t)}) \quad (20)$$

$$y^{(2)}(m) = \sum_{t=1}^m (\alpha x^{(1)}(t + \Delta t_1) + x^{(2)}(t + \Delta t_2) - \overline{\alpha x^{(1)}(t + \Delta t_1)} - \overline{x^{(2)}(t + \Delta t_2)}) \quad (21)$$

To not affect the overall characteristics of $\{x^{(2)}(t)\}$ too much, the introduction parameter α must be very small.

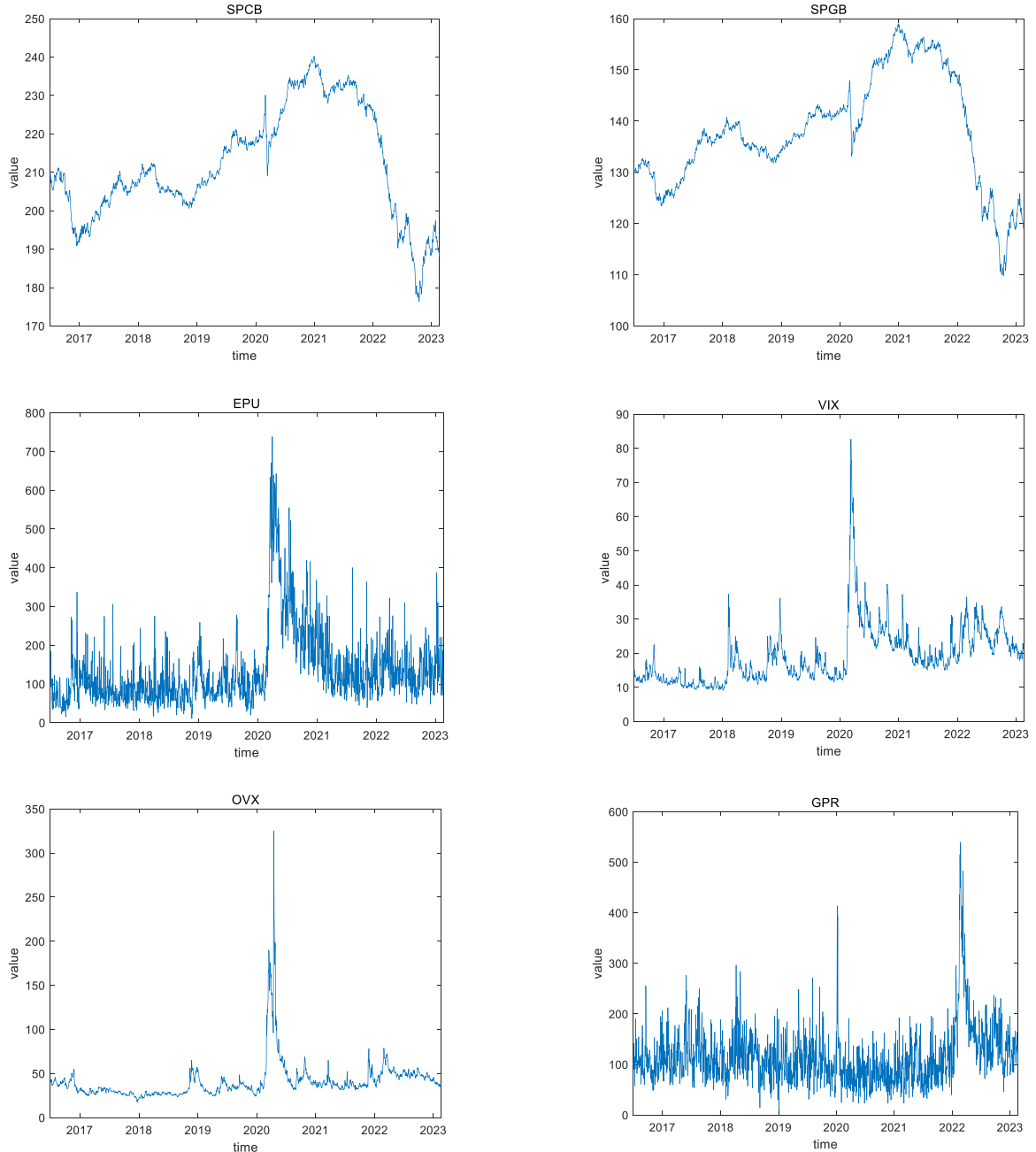


Fig. 1. Time series plot of uncertainty indices and bond market prices.

$$\begin{cases} \alpha = \frac{\sum_{t=1}^m (x^{(2)}(t + \Delta t_2))^2}{(M - \Delta t_1) \sum_{t=1}^m (x^{(1)}(t + \Delta t))^2}, \Delta t_1 \geq \Delta t_2, m = M - \Delta t_1 \\ \alpha = \frac{\sum_{t=1}^m (x^{(2)}(t + \Delta t))^2}{(M - \Delta t_2) \sum_{t=1}^m (x^{(1)}(t + \Delta t))^2}, \Delta t_1 \leq \Delta t_2, m = M - \Delta t_2 \end{cases} \quad (22)$$

The remaining steps are consistent with MF-ADCCA steps 2–4, from which we obtain the time-lagged multifractal correlation between the two time series.

4. Empirical analysis

4.1. Data selection and preliminary statistical test

To examine the relationship between uncertainties and bond markets from the multifractal perspective, this study selects four uncertainty indices and the conventional and green bond prices, encompassing the economic policy uncertainty (EPU) index and geopolitical risk index (GPR) from <http://www.policyuncertainty.com/>, the market volatility index (VIX) and crude oil volatility index (OVX) from <https://investing.com/> and the conventional bond prices (SPCB) and green bond prices (SPGB) from <https://www.spglobal.com/>. After data preprocessing, the sample time interval covers July 5, 2016 to February 27, 2023, with a total of 1568 samples.

Fig. 1 presents the time series of the raw data, revealing that the market prices of conventional and green bonds exhibit an upward trend from 2016 to 2020. The two types of bonds plunged in price due to the COVID-19 pandemic, while the values of EPU, VIX, and OVX indices peaked during the COVID-19 outbreak. The outbreak of the Russia–Ukraine war is associated with a large GPR index value around 2022. For follow-up analysis, this study examines bond market returns and uncertainty index changes using the specific calculation formula $r_t = \ln(p_t/p_{t-1})$.

Table 1 presents the descriptive statistics of bond market returns and uncertainty index changes, revealing a large standard deviation of the changes in EPU and GPR indices, which indicates that the series are more volatile. The skewness of all series is far from the zero skewness of the normal distribution, demonstrating that the series are skewed. Kurtosis values are greater than those of normal distribution 3, and Jarque–Bera (J–B) statistics reject the normal distribution at a 1 % significance level.

The original hypothesis of the modified Dickey–Fuller (DF)-GLS (Elliott et al., 1992) test is that the series is nonstationary, and this test rejects the original hypothesis that the time series is stationary, while the original hypothesis of the Kwiatkowski, Phillips, Schmidt, and Shin (KPSS; Kwiatkowski et al., 1992) test is that the series is stationary. This test accepts the original hypothesis that the time series is stationary and considers the constant and linear trend for both tests. As shown in Table 2, under the DF-GLS test, the original hypothesis of the existence of unit root is rejected at the 1 % significance level. For the KPSS test, the original hypothesis of the stationarity of the selected series could not be rejected; therefore, DF-GLS and KPSS tests demonstrate that the test series are stationary.

Before conducting the multifractal analysis, we conduct the Qcc intercorrelation test between uncertainty index changes and bond market returns with degrees of freedom (m) from 1 to 300. Fig. 2 reveals that as the degrees of freedom rise, the intercorrelation test statistics between uncertainties and bond markets are above the 5 % significance level, rejecting the original hypothesis of no intercorrelation between uncertainties and bond markets. Therefore, a long-term memory correlation is evident between uncertainties and bond markets, confirming that multifractal correlation analysis can be conducted.

4.2. Asymmetric multifractal detrended cross-correlation analysis

This subsection uses MF-ADCCA to investigate the correlation between uncertainty index changes and bond market returns, applying time scale s as $10 \leq s \leq (N/5)$, fitting polynomials of order $m = 2$, q from -4 –4.

4.2.1. Multifractal relationships between uncertainty and bond markets under different trends

Figs. 3 and 4 present the double logarithm graph of the fluctuation function against the time scale between the uncertainty index changes, and bond returns when $q = 2$. As can be seen from Figs. 3 and 4, for smaller time scales s , no significant asymmetry is found. As the time scale s increases, the degree of asymmetry between the change in the uncertainty index and bond market returns increases. For

Table 1
Descriptive statistics.

	SPCB	SPGB	EPU	VIX	OVX	GPR
Min	− 2.1210	− 2.4099	− 183.265	− 29.9831	− 62.2251	− 299.6048
Max	2.0775	2.2717	230.3838	76.8245	85.7700	234.5083
Mean	− 0.0066	− 0.0058	0.02400	0.0189	− 0.0042	0.0015
Std	0.3557	0.3827	47.8765	8.2625	6.8577	45.4145
Skewness	− 0.3975	− 0.4280	0.1680	1.4622	2.0119	− 0.1192
Kurtosis	7.4537	8.3632	4.0245	11.3175	32.5452	4.9188
J–B test	1336.3***	1925.9***	75.89***	5075.26***	5,8052***	244.11***

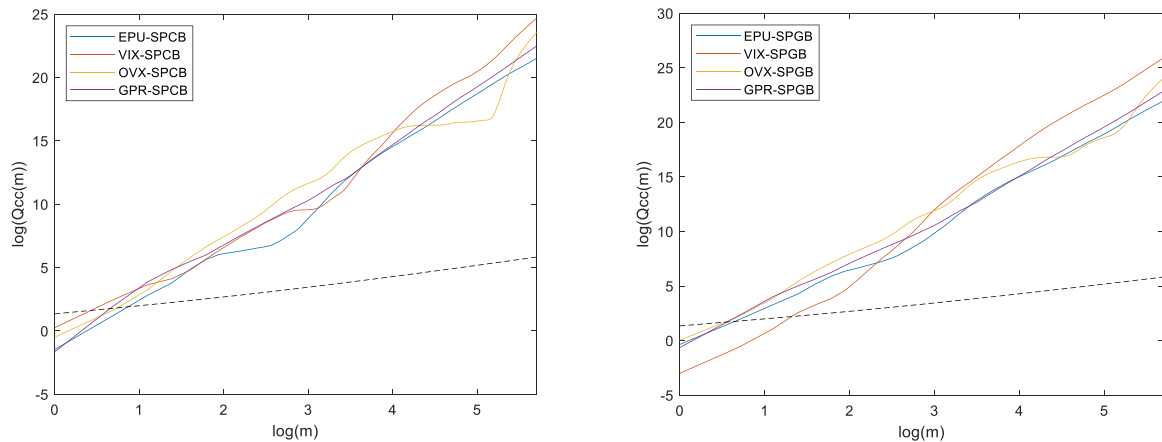
Note: * * * indicates significance at the 1 % level.

Table 2

Unit-root test of uncertainty index changes and bond market returns.

	DF-GLS		KPSS	
	Constant	Trend	Constant	Trend
SPCB	− 10.6878***	− 10.4496***	0.3811	0.1770
SPGB	− 5.9437***	− 7.8304***	0.4831	0.1428
EPU	− 0.8390***	− 2.2072***	0.0052	0.0053
VIX	− 4.8296***	− 7.8522***	0.0139	0.0140
OVX	− 5.2087***	− 8.2915***	0.0218	0.0217
GPR	− 5.3002***	− 8.7802***	0.0038	0.0039

Note: * ** indicates significance at the 1 % level.

**Fig. 2.** Statistical test of Qcc intercorrelation between uncertainty and bond markets. Note: The dashed line denotes the critical value of the chi-square statistic at a 5 % significance level.

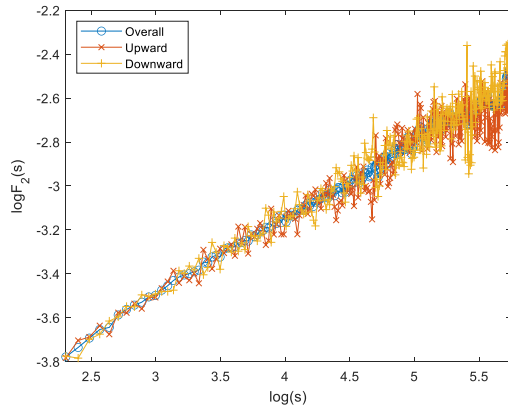
the uncertainty index, the asymmetry between the change of market volatility index, changes of crude oil volatility index and bond market return is more obvious.

Using Eqs. (12)–(14), the Hurst exponent between the two variables can be calculated for different trends. Tables 3 and 4 present the values of the Hurst exponent between uncertainty index changes and conventional and green bond market returns at the time of different trends in uncertainty index changes. Tables 3 and 4 reveal that regardless of the trend (overall, upward, and downward trends), $H_{12}(2)$, $H_{12}^+(2)$, $H_{12}^-(2)$ are less than 0.5, indicating that changes in EPU and geopolitical risk indices have an antipersistent relationship with bond market returns, and increased EPU or geopolitical risk decreases the prices of both types of bond markets. When the two uncertainty indices exhibit an upward or downward trend, the changes are inversely correlated with long-term bond market returns. This demonstrates that when risks in economic and geopolitical fields are significant, bond markets will be negatively affected by the market volatility. In contrast, when the risks in the economic and geopolitical fields are minimal, bond markets are more prosperous and price rise steadily. Therefore, it is essential to guard against circumstances in which EPU and geopolitical risks increase for both bond markets. When an upward trend ($H_{12}^+ < 0.5$) occurs, the relationship between changes in market volatility and crude oil volatility indices and bond market returns is antipersistent. When overall and downward trends ($H_{12}(2)$, $H_{12}^-(2) > 0.5$) occur, the relationship between the changes in market and crude oil volatility indices and bond market returns remains persistent. This means that when market and crude oil volatility indices increase, stock or the crude oil markets are more volatile. Investors and market stakeholders diversify their exposure to stock or energy markets by including a higher proportion of bonds in their portfolios to diminish exposure to stock or energy market volatility.

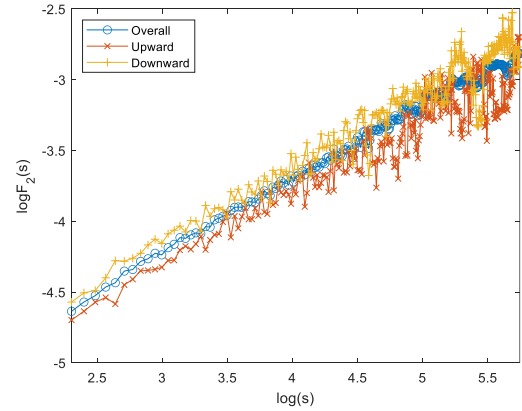
$\Delta H_{12}(2)$ measures the difference between the Hurst exponent for upward and downward trends. Table 3 indicate that the difference between upward and downward trends in crude oil market volatility is the largest. This is attributable to the high volatility of the oil market, the increasing degree of financial orientation in the contemporary market, and the global economy's high dependence on energy. Because of these series of factors, the crude oil market has a stronger impact on other markets. Meanwhile, bond markets can hedge the risks from the oil market, resulting in a highly heterogeneous multifractal correlation relationship between the crude oil volatility index and the bond market under different trends. For other uncertainties, the difference between uncertainty and the green bond market under different trends is generally greater than that between uncertainty and the conventional bond market. This is attributed to green bonds having lower volatility than conventional bonds and becoming a more popular diversified investment tool.

4.2.2. Multifractal relationships between uncertainty and bond markets under large and small fluctuations

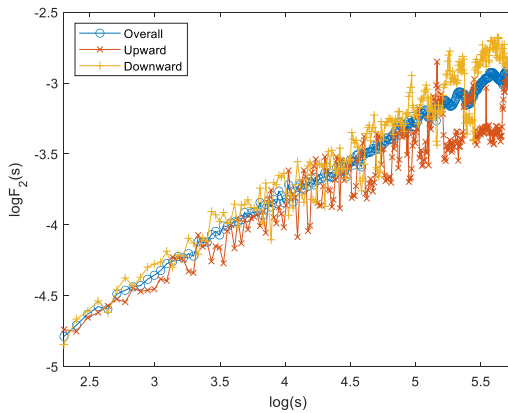
As shown in Figs. 5 and 6, the Hurst exponent of uncertainties and bond markets varies with q under the three trends, confirming



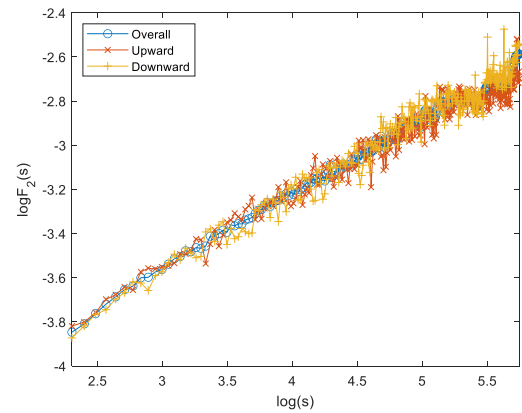
(a) EPU-SPCB



(b) VIX-SPCB



(c) OVX-SPCB

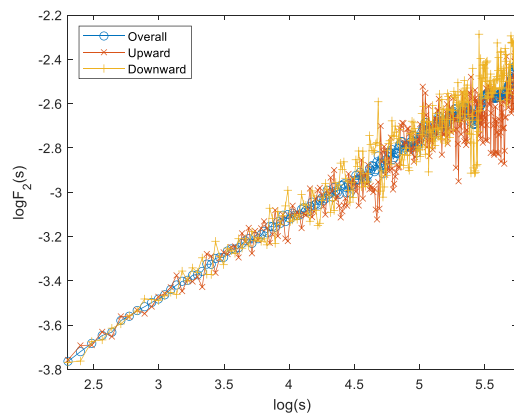


(d) GPR-SPCB

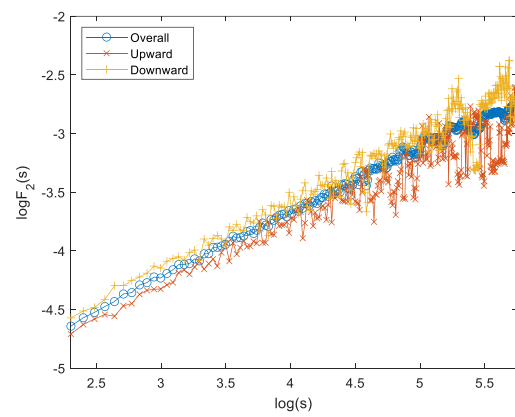
Fig. 3. Dividing the trends in uncertainty index changes, the double logarithm graph of the fluctuation function against the time scale between the uncertainty index changes, and conventional bond returns.

the existence of multifractal relationships between uncertainties and bond markets. Regardless of the change trends in EPU and geopolitical risk indices (overall, upward, and downward trends), the changes in these indices are antipersistent across conventional and green bond markets under small ($q < 0$) and large fluctuations ($q > 0$), indicating that regardless of whether the fluctuations are small or large, EPU or geopolitical risk are inversely correlated with both bond markets in the long run. For conventional bonds, changes in market and crude oil volatility indices are persistent with conventional bond market returns, regardless of the trends in the small fluctuation scenario. In the large fluctuation scenario, changes in the market volatility index are inverse persistent with conventional bond returns in all three trends, and changes in the crude oil volatility index have an antipersistent relationship with conventional bond returns in the downward trend and an antipersistent relationship with conventional bond market returns in the upward trend. For green bonds, changes in market and crude oil volatility indices are persistent with green bond returns in the small fluctuation scenario. In the large fluctuation scenario, the two uncertainty indices are persistent with green bond returns in the downward trend and antipersistent with green bond returns in the upward trend. The results reveal that market and crude oil volatility indices have an antipersistent relationship with both bond markets in large fluctuation scenarios and an antipersistent relationship in small fluctuation scenarios because volatility is more intense in the crude oil and stock markets when large fluctuations occur compared with small fluctuations. Although investors and market stakeholders invest more in bonds to hedge risks, the high volatility of crude oil and stock markets transmits risk to bond markets, lowering bond market prices. In times of small fluctuation in stock and crude oil markets, some assets are transferred from stocks or oil to bonds, causing bond prices to rise. The difference between green and conventional bonds when confronting uncertainty is primarily attributed to financial markets' varying volatility and returns in the face of uncertainty risk. The ongoing global quest for green development and the continued increase in green market investment have made green markets less exposed to risk shocks.

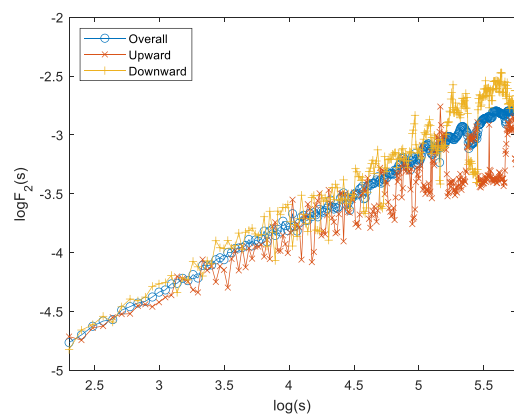
Fig. 7 shows that the value of $\Delta H_{12}(q)$ with q between changes in geopolitical risk and EPU indices with bond market returns is small (i.e., a small degree of asymmetry). The absolute value of $\Delta H_{12}(q)$ with q between changes of the crude oil volatility index and bond market returns is larger (i.e., a higher degree of asymmetry). Moreover, the negative value of $\Delta H_{12}(q)$ indicates that the relationship between changes in the uncertainty index and bond market returns is more persistent when the uncertainty index is in a



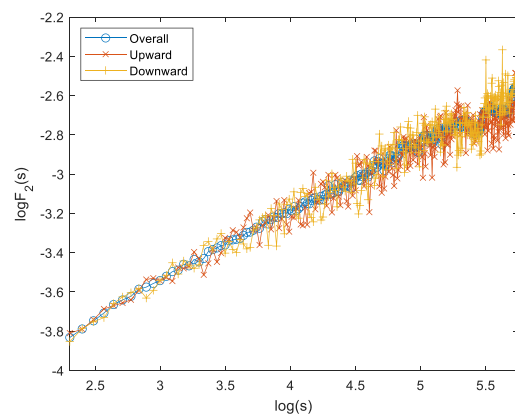
(a) EPU-SPGB



(b) VIX-SPGB



(c) OVX-SPGB



(d) GPR-SPGB

Fig. 4. Dividing the trends in uncertainty index changes, the double logarithm graph of the fluctuation function against the time scale between the uncertainty index changes, and green bond returns.

Table 3

The Hurst exponent of uncertainty index changes and conventional bond returns under different uncertainty index trends.

	EPU-SPCB	VIX-SPCB	OVX-SPCB	GPR-SPCB
$H_{12}(2)$	0.3433	0.5069	0.5279	0.3244
$H_{12}^+(2)$	0.3291	0.4845	0.4401	0.3068
$H_{12}^-(2)$	0.3563	0.5076	0.5653	0.3425
$\Delta H_{12}(2)$	-0.0272	-0.0231	-0.1252	-0.0357

Table 4

The Hurst exponent of uncertainty index changes and green bond returns under different uncertainty index trends.

	EPU-SPGB	VIX-SPGB	OVX-SPGB	GPR-SPGB
$H_{12}(2)$	0.3580	0.5343	0.5762	0.3348
$H_{12}^+(2)$	0.3339	0.4851	0.4443	0.3170
$H_{12}^-(2)$	0.3755	0.5443	0.6220	0.3502
$\Delta H_{12}(2)$	-0.0416	-0.0592	-0.1777	-0.0332

downward trend than when it is in an upward trend. This is primarily attributed to the fact that when the uncertainty index is in an uptrend, it has a stronger inverse correlation with bond markets and bond markets are more volatile.

4.2.3. Multifractal spectrum

The multifractal spectrum can also measure the multifractal characteristics between two variables. As shown in Fig. 8, the

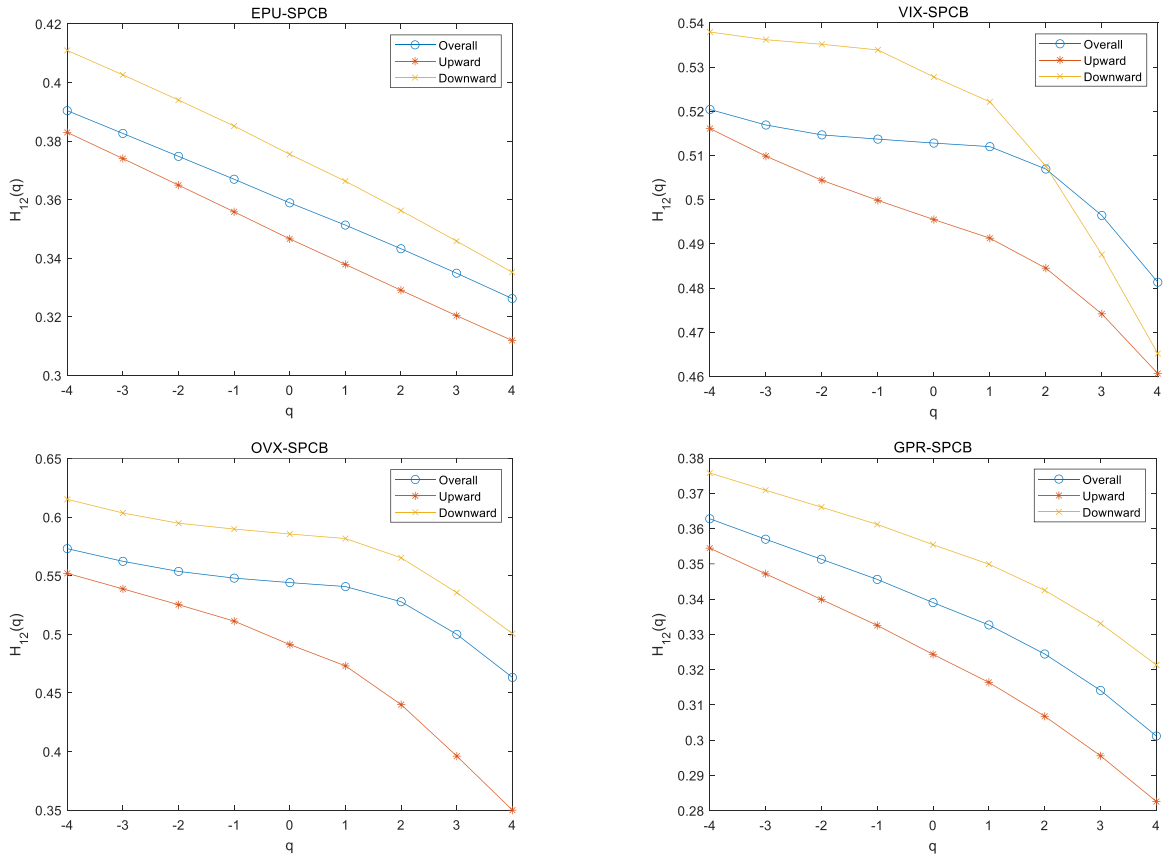


Fig. 5. Dividing the trends in uncertainty index changes, the value of the general Hurst exponent with q of the uncertainty index changes, and conventional bond returns.

multifractal spectrum is not concentrated at one point but has a smooth curved shape, demonstrating that uncertainties and bond markets previously exhibited multifractal characteristics under different trends. Furthermore, the images are all left-skewed, suggesting that large fluctuations are the primary cause of the multifractal characteristics between uncertainties and bond markets under different trends.

4.3. MF-DCCA method based on risk transmission

The previous section confirmed the existence of multifractal cross-correlation between uncertainties and bond markets and described the multifractal characteristics of this cross-correlation from various perspectives. To further examine the multifractal correlation between these factors and explore the risk relationship between uncertainties and bond markets, we employ MF-DCCA based on the risk conduction method to investigate the long memory correlation between uncertainties and bond markets under different time lags to rationalize the characteristics of risk transmission between them. Finally, we compare the heterogeneity of risk transmission characteristics between uncertainties and the two types of bond markets using different time lags.

4.3.1. Risk transmission relationship between uncertainty and bond markets

The profile series of uncertainty index changes and bond market returns using different time lags are respectively obtained using the following equations:

$$y^{(1)}(m) = \sum_{t=1}^m (x^{(1)}(t) - \overline{x^{(1)}(t)})$$

$$y^{(2)}(m) = \sum_{t=1}^m (\alpha x^{(1)}(t + \Delta t_1) + x^{(2)}(t + \Delta t_2) - \overline{\alpha x^{(1)}(t + \Delta t_1)} - \overline{x^{(2)}(t + \Delta t_2)})$$

According to [Cao et al. \(2017\)](#), when the relationship between two sequences is completely linear, then it is unnecessary to make Δt_1 equal Δt_2 because the correlation between them obtained in this way is unstable. The previous section demonstrates a nonlinear multifractal correlation between uncertainties and bond markets. For the convenience of the period, we employ the delay time Δt_1 equals Δt_2 , with a value from 1 to 20 (20 days trading time is about one month).

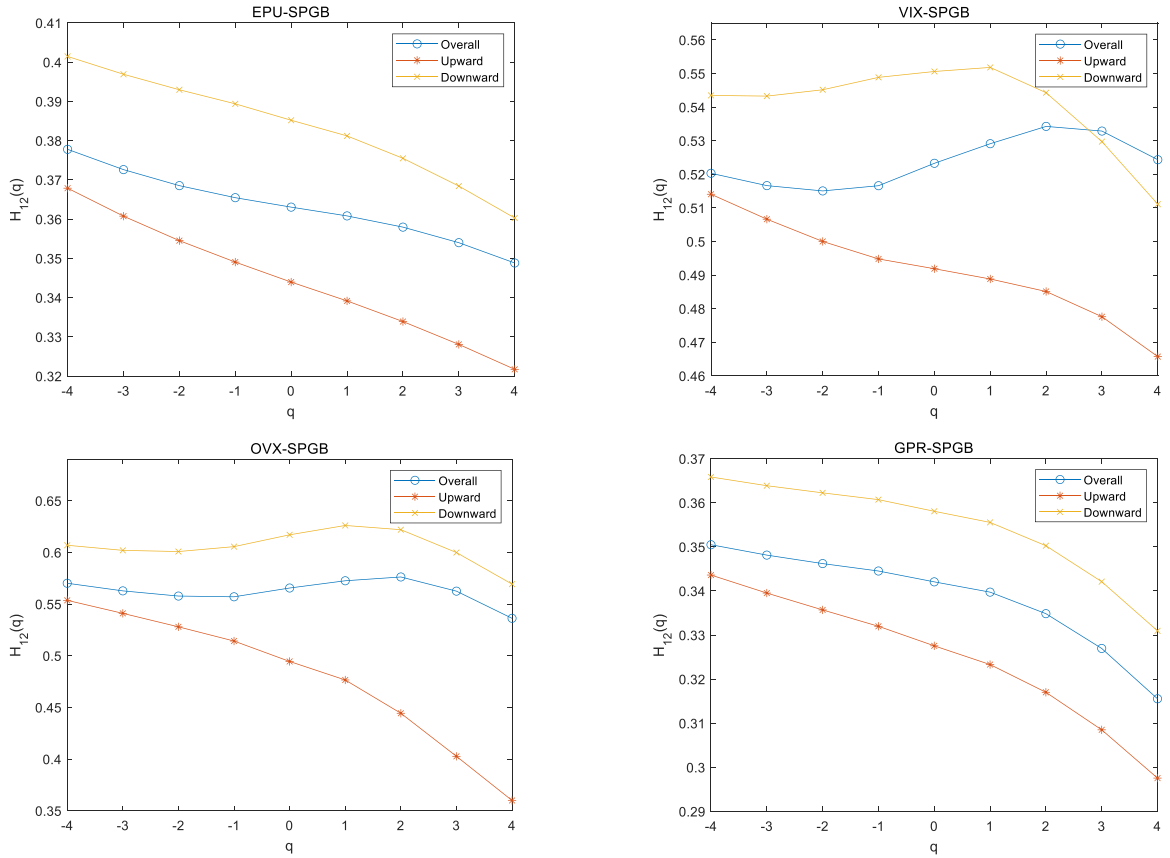


Fig. 6. Dividing the trends in uncertainty index changes, the value of the general Hurst exponent with q of uncertainty index changes, and green bond returns.

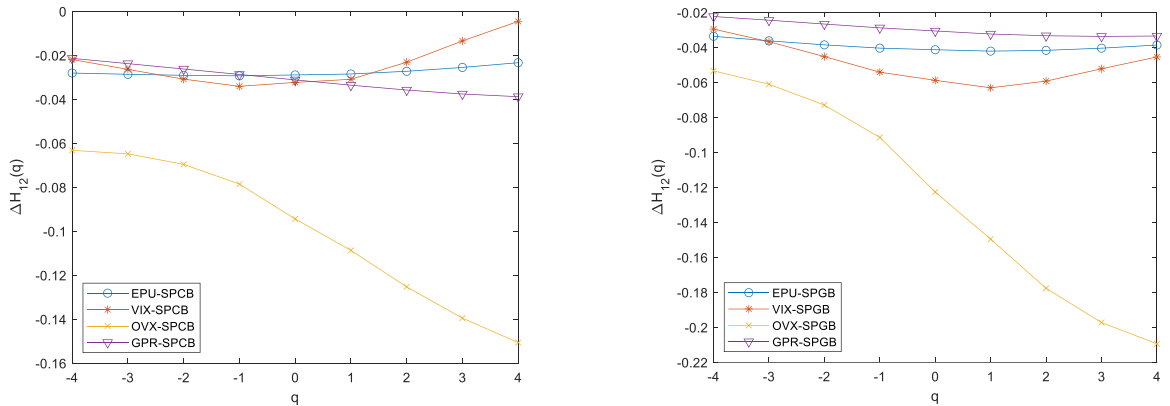
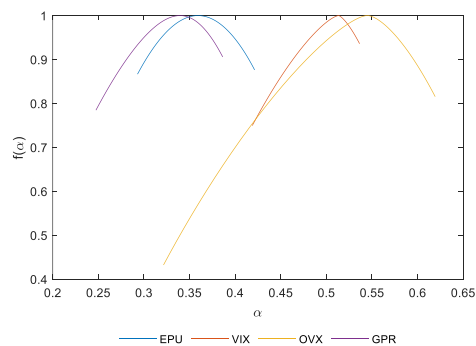
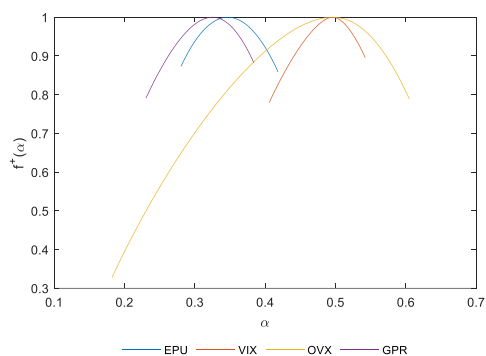


Fig. 7. Dividing the trends in uncertainty index changes, the $\Delta H_{12}(q)$ value versus q between the uncertainty index changes, and bond market returns.

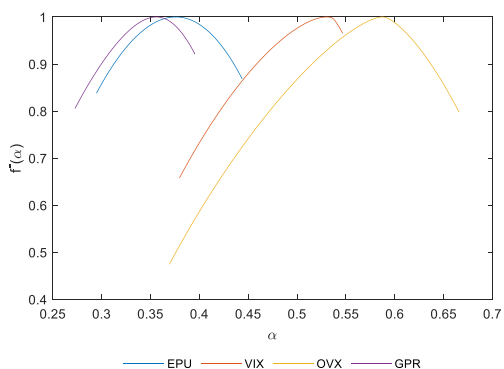
As shown in Figs. 9 and 10, when the bond market lags, an antipersistence relationship between changes in EPU and geopolitical risk indices emerges with the two different types of bond market returns when the bond market lags by 1–20. This means that the increased risk of economic and geopolitical uncertainty with a lag of 1–20 will inevitably lower bond market prices; therefore, it is necessary to closely monitor the effects of economic and geopolitical uncertainty and the continuing impact of these effects on bond markets. A persistent relationship is evident between the changes in the market volatility index and bond market returns under a short time lag. When the lag time is extended, an antipersistent correlation emerges between changes in the market volatility index and conventional bond market returns, and a persistent correlation emerges between changes in the market volatility index and green bond



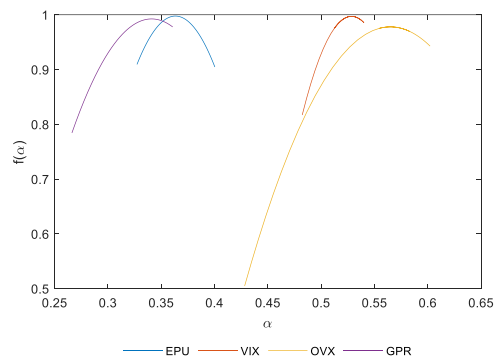
Uncertainty-SPCB (overall trend)



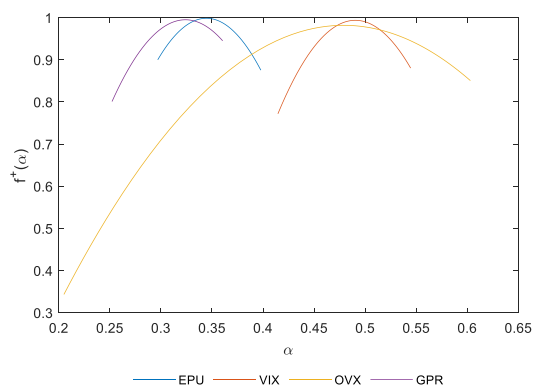
Uncertainty-SPCB (upward trend)



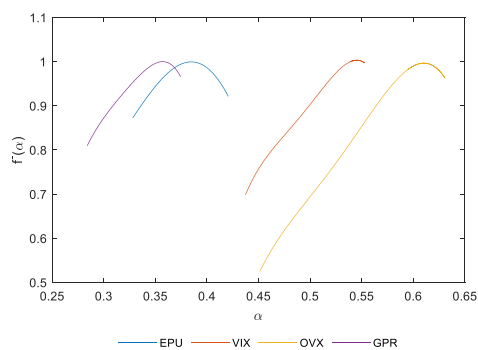
Uncertainty-SPCB (downward trend)



Uncertainty-SPGB (overall trend)



Uncertainty-SPGB (upward trend)



Uncertainty-SPGB (downward trend)

Fig. 8. Multifractal spectrum of uncertainty and bond markets.

market returns. When the bond market is lagged by 1–20, a persistent correlation emerges between crude oil volatility index changes and bond market returns. Market and crude oil volatility indices have a predominantly persistent relationship with bond markets when the lag is 1–20, indicating that when stock and crude oil markets are volatile, in a long lag period, investors will transfer a portion of their assets from stocks or crude oil to bonds, raising bond prices.

Figs. 11 and 12 show that when the changes in the uncertainty index lag, changes in EPU and geopolitical risk indices lag by 1–20,

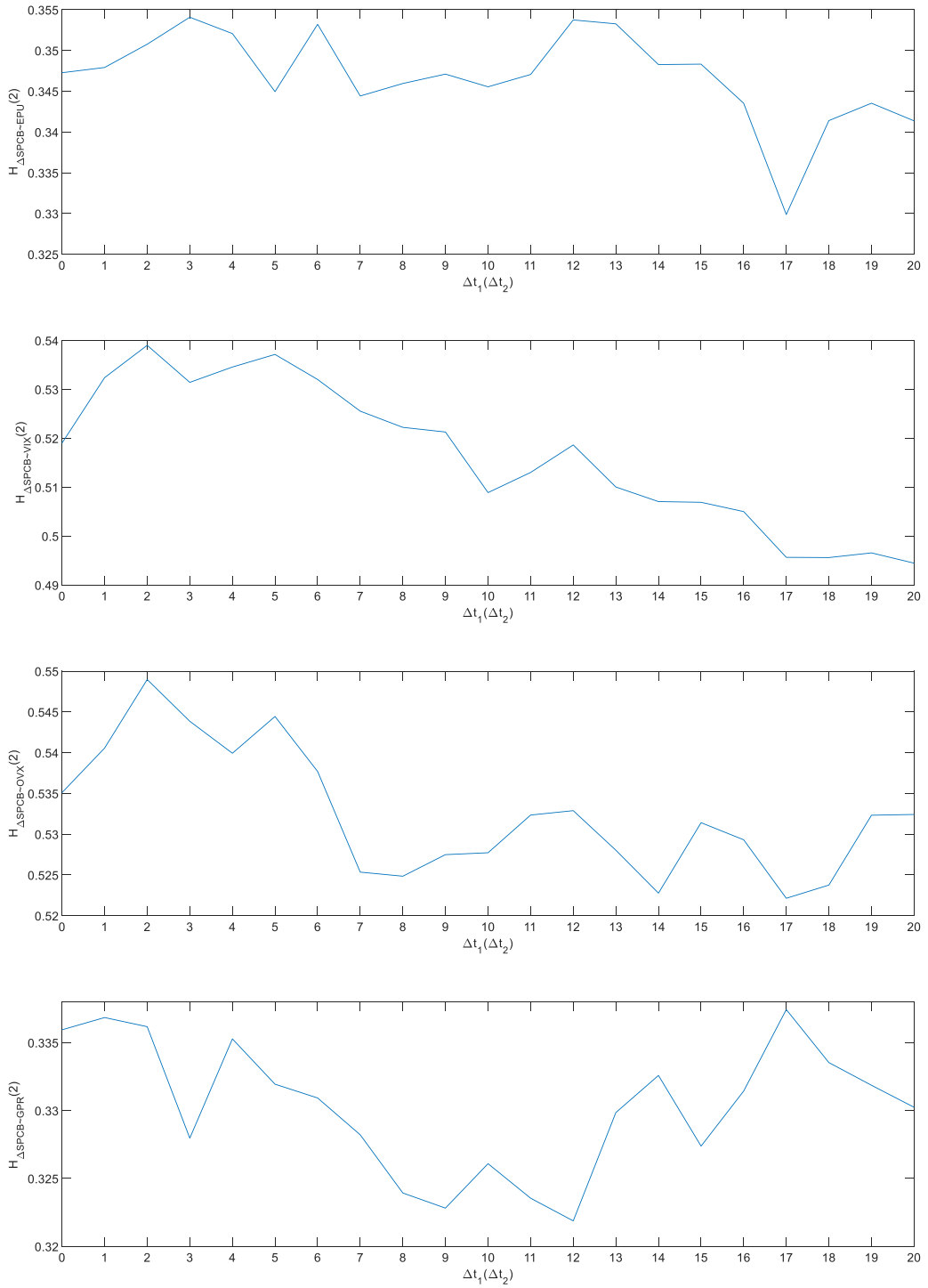


Fig. 9. Hurst values of the uncertainty index changes and conventional bond returns at $q = 2$ with different time lags in the conventional bond market.

exhibiting an antipersistent relationship with the two different types of bond market returns. This is because when EPU and geopolitical risks increase sharply, sharp falls in bond market prices are generally caused by economic crises and geopolitical risks. Bond market decline will transmit bond market risks to other financial markets, which will have a further negative impact on the economy and deepen the long-term negative correlation between EPU, geopolitical risks, and bond markets. Changes in the market volatility index have a persistent correlation with green and conventional bond markets under a short time lag. With a long lag time, an

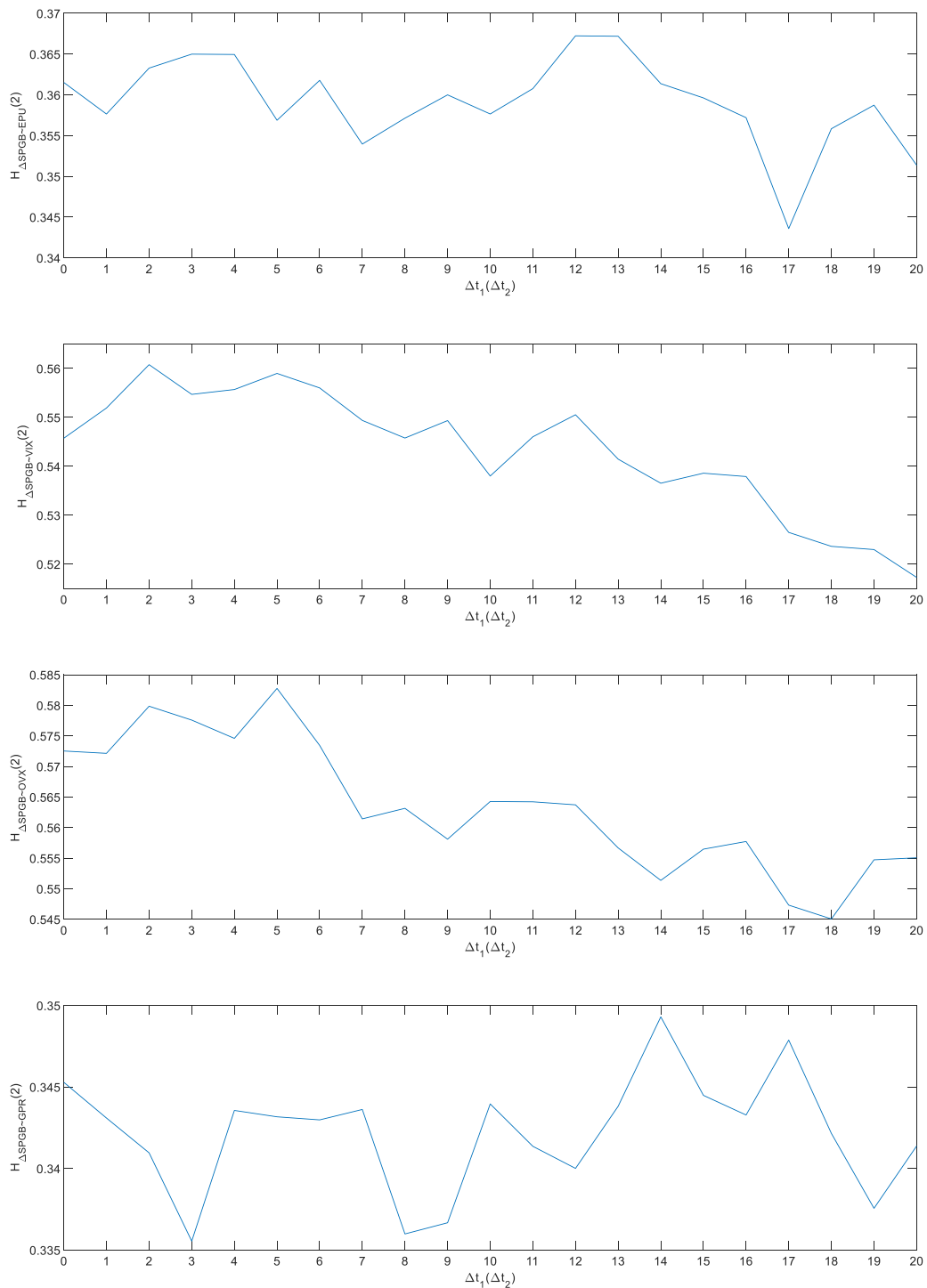


Fig. 10. Hurst values of uncertainty index changes and green bond returns at $q = 2$ with different time lags in the green bond market.

antipersistence relationship emerges with conventional bond market returns and a persistent relationship with green bond market returns. This is primarily attributed to the transfer of higher-risk assets to lower-risk bonds when a market crisis occurs, and the hedging ability of green bonds is generally stronger than conventional bonds. As a result, when a crisis arises, a greater shift in assets to green bonds occurs; therefore, green bonds generally have stronger persistence than conventional bonds with a lagged market volatility index. When the crude oil volatility index changes with a lag of 1–20, a persistent relationship occurs with the two bond markets' returns, which is attributable to bonds being a better hedge against crude oil.

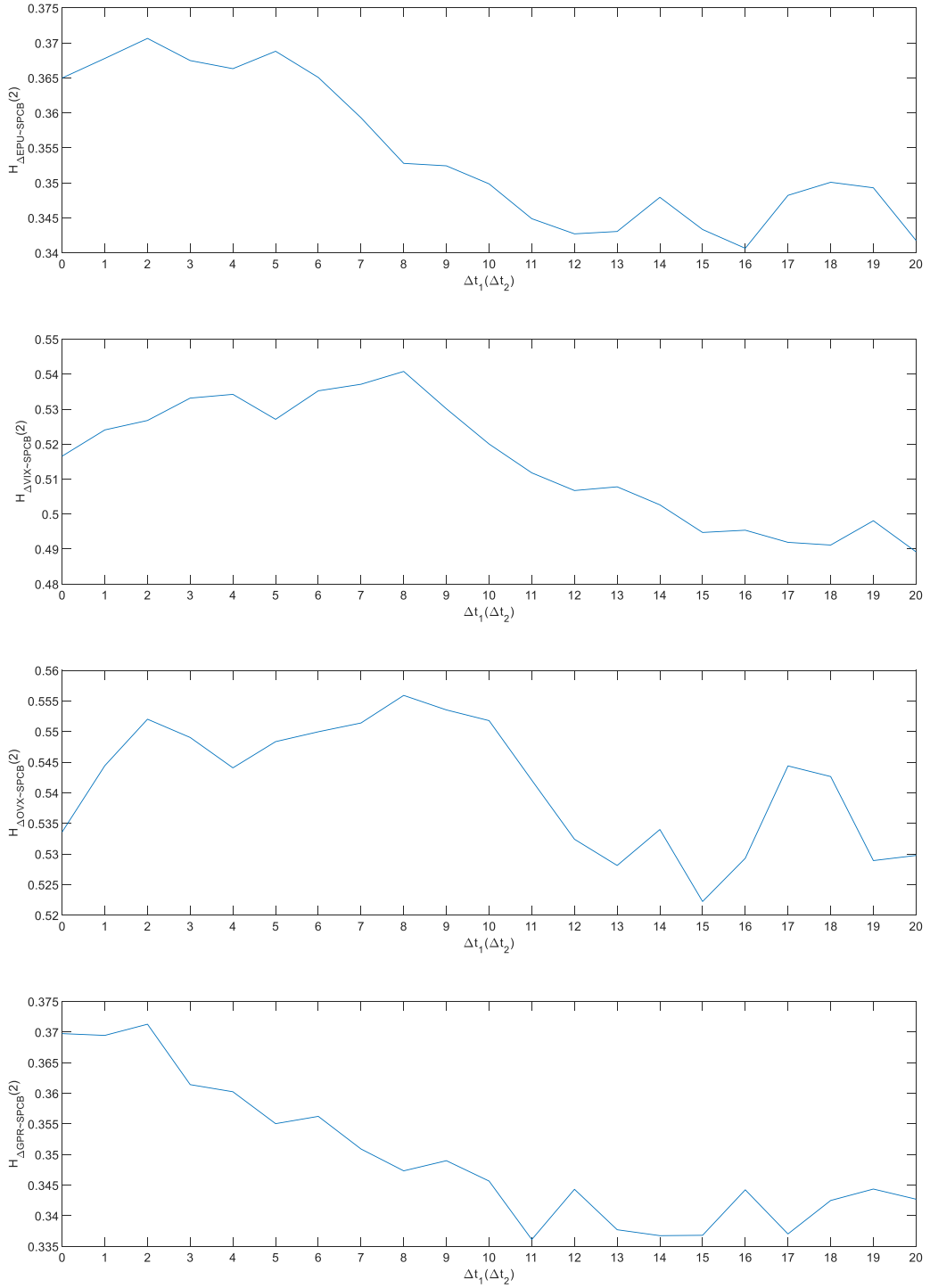


Fig. 11. Hurst values of the uncertainty index changes and conventional bond returns at $q = 2$ with different time lags in uncertainty index changes.

4.3.2. Heterogeneity in the risk transmission relationship between uncertainties and green and conventional bond markets

To compare the differences in uncertainties' risk transmission on green and conventional bond markets, we calculate the difference of the Hurst exponent between the changes in uncertainty indices and the two different types of bond market returns with different time lags (denoted as $H^{\Delta T}(2)$) as follows:

$$H_{\text{EPU} \sim \text{Bond}}^{\Delta T}(2) = H_{\Delta \text{SPCB} \sim \text{EPU}}(2) - H_{\Delta \text{SPGB} \sim \text{EPU}}(2)$$

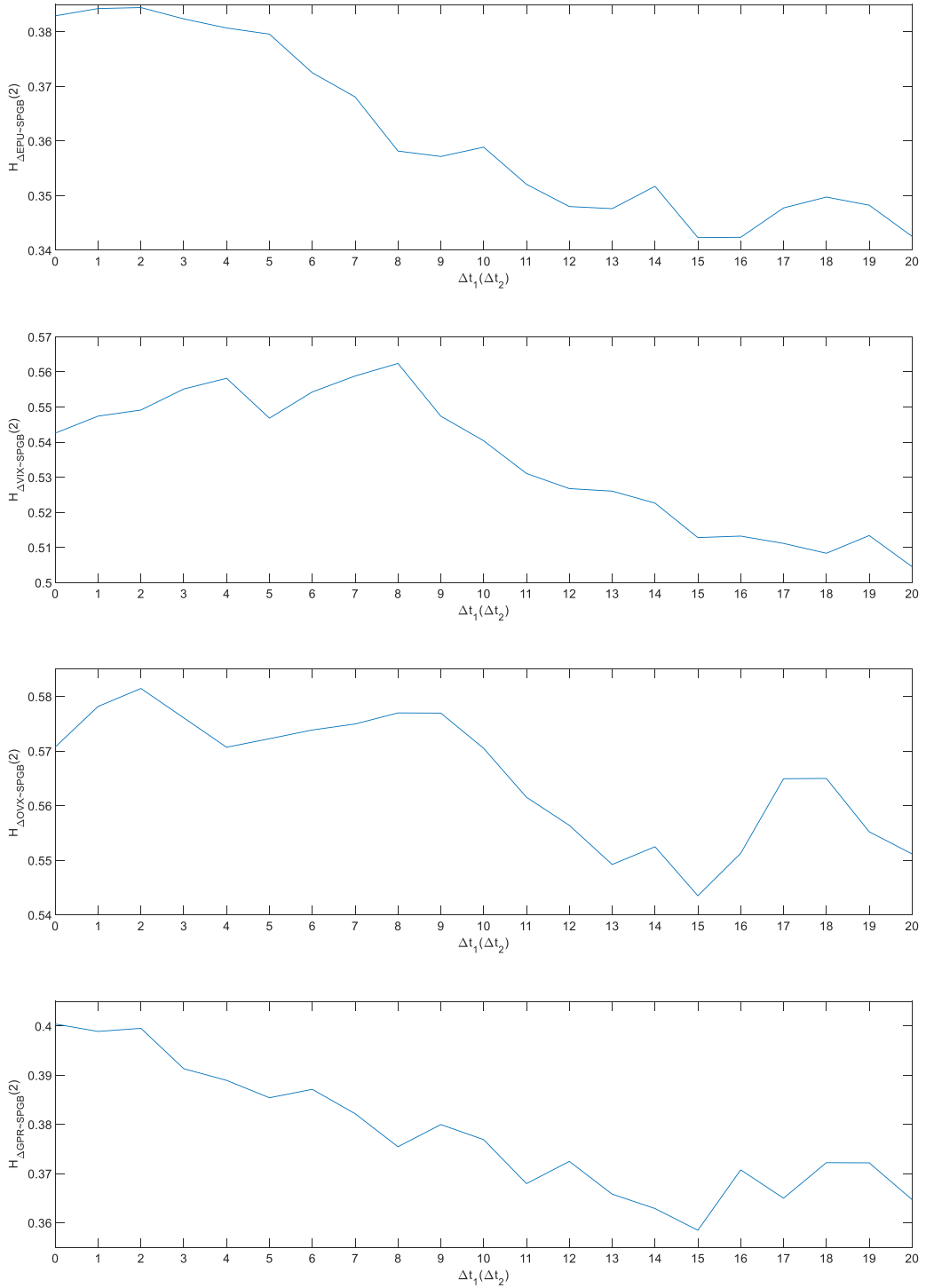


Fig. 12. Hurst values of uncertainty index changes and green bond returns at $q = 2$ with different time lags in uncertainty index changes.

$$H_{Bond \sim EPU}^{\Delta T}(2) = H_{\Delta EPU \sim SPCB}(2) - H_{\Delta EPU \sim SPGB}(2)$$

$$H_{VIX \sim Bond}^{\Delta T}(2) = H_{\Delta SPCB \sim VIX}(2) - H_{\Delta SPGB \sim VIX}(2)$$

$$H_{Bond \sim VIX}^{\Delta T}(2) = H_{\Delta VIX \sim SPCB}(2) - H_{\Delta VIX \sim SPGB}(2)$$

$$H_{OVX \sim Bond}^{\Delta T}(2) = H_{\Delta SPCB \sim OVX}(2) - H_{\Delta SPGB \sim OVX}(2)$$

$$H_{Bond \sim OVX}^{\Delta T}(2) = H_{\Delta OVX \sim SPCB}(2) - H_{\Delta OVX \sim SPGB}(2)$$

$$H_{GPR \sim Bond}^{\Delta T}(2) = H_{\Delta SPCB \sim GPR}(2) - H_{\Delta SPGB \sim GPR}(2)$$

$$H_{Bond \sim GPR}^{\Delta T}(2) = H_{\Delta GPR \sim SPCB}(2) - H_{\Delta GPR \sim SPGB}(2)$$

Fig. 13 reveals that when bond market returns lag, uncertainties and the green bond market have stronger persistence than the conventional bond market. EPU and geopolitical risk exhibit minimal differences in risk transmission with the two bond markets, while market and crude oil volatility indices cause a large difference between them. This indicates that when uncertainty rises, compared with conventional bonds, green bonds can better maintain the upward trend, or the downward trend is weaker, indicating that green bonds have stronger risk-aversion capabilities compared with conventional bonds against various uncertainties. The difference between EPU and geopolitical risk in risk transmission is small because the events that are correlated with significant growth in geopolitical risk and EPU are often accompanied by economic turmoil, and conventional bond market is more volatile. However, fluctuations in crude oil and the market cause a considerable difference in risk transmission, which is attributed to green bonds generally serving as better safe-haven assets than conventional bonds. Fig. 14 demonstrates that when the uncertainty index lags, a minimal difference is evident between green and conventional bond markets on the time delay of EPU, and the difference between them on the time delay of the geopolitical risk index is large.

To compare the differences in risk transmission directions between uncertainty index changes and bond market returns, we next calculate the differences between the Hurst exponent in places where the uncertainty index and bond market lag. As shown in Figs. 15 and 16, the risk transmission effect of the market volatility index on the bond markets is stronger than transmission from bond markets to the market volatility index under short- and long-term time lags. This is because a high market volatility index is often observed in the context of dramatic market volatility. The bond market has a stronger risk transmission effect on EPU under the short-term time lag, and the direction of this transmission effect changes under the long-term time lag. This indicates that EPU takes time to affect bond markets. The transmission effect of the bond market on the geopolitical risk index is stronger than that of the geopolitical risk index on the bond market.

$$H_{EPU \sim SPCB}^{\Delta T}(2) = H_{\Delta SPCB \sim EPU}(2) - H_{\Delta EPU \sim SPCB}(2)$$

$$H_{EPU \sim SPGB}^{\Delta T}(2) = H_{\Delta SPGB \sim EPU}(2) - H_{\Delta EPU \sim SPGB}(2)$$

$$H_{VIX \sim SPCB}^{\Delta T}(2) = H_{\Delta SPCB \sim VIX}(2) - H_{\Delta VIX \sim SPCB}(2)$$

$$H_{VIX \sim SPGB}^{\Delta T}(2) = H_{\Delta SPGB \sim VIX}(2) - H_{\Delta VIX \sim SPGB}(2)$$

$$H_{OVX \sim SPCB}^{\Delta T}(2) = H_{\Delta SPCB \sim OVX}(2) - H_{\Delta OVX \sim SPCB}(2)$$

$$H_{OVX \sim SPGB}^{\Delta T}(2) = H_{\Delta SPGB \sim OVX}(2) - H_{\Delta OVX \sim SPGB}(2)$$

$$H_{GPR \sim SPCB}^{\Delta T}(2) = H_{\Delta SPCB \sim GPR}(2) - H_{\Delta GPR \sim SPCB}(2)$$

$$H_{GPR \sim SPGB}^{\Delta T}(2) = H_{\Delta SPGB \sim GPR}(2) - H_{\Delta GPR \sim SPGB}(2)$$

5. Discussion

5.1. Why do green bonds differ from conventional bonds?

The primary difference between green and conventional bonds is the use of proceeds. Green bonds are specifically used to finance environmental protection projects, often featuring long-term sustainable development such as renewable energy and clean transportation (Ren et al., 2023; Cao et al., 2024). Because such projects are closely related to environmental risks and the challenges of climate change, investing in green bonds seeks financial returns while emphasizing social benefits and environmental protection. During rare disasters such as financial crises, extreme climate events or pandemics, green bonds may exhibit stronger resilience related to long-term investment characteristics and positive environmental impact (Dong et al., 2023; Silva et al., 2024). Additionally, green bond issuers typically have higher credit ratings, which is attributable to stricter review standards set by regulatory authorities for issuing green bonds (Wang and Shen, 2024; Li et al., 2025). Higher credit ratings imply lower default, which increases the safety of green bonds. Furthermore, green bonds often enjoy support from governments and regulatory authorities, including tax incentives and subsidies (Wang and Liu, 2024). This support helps reduce issuance costs and enhances investor confidence, further strengthening the risk resilience of green bonds. Finally, with the growing popularity of environmental, social, and governance investment, green bonds are becoming increasingly favored by investors (Mohammed et al., 2024), which may provide additional support for green bonds during market turmoil. However, compared with the conventional bond market, the green bond market is smaller in scale and may

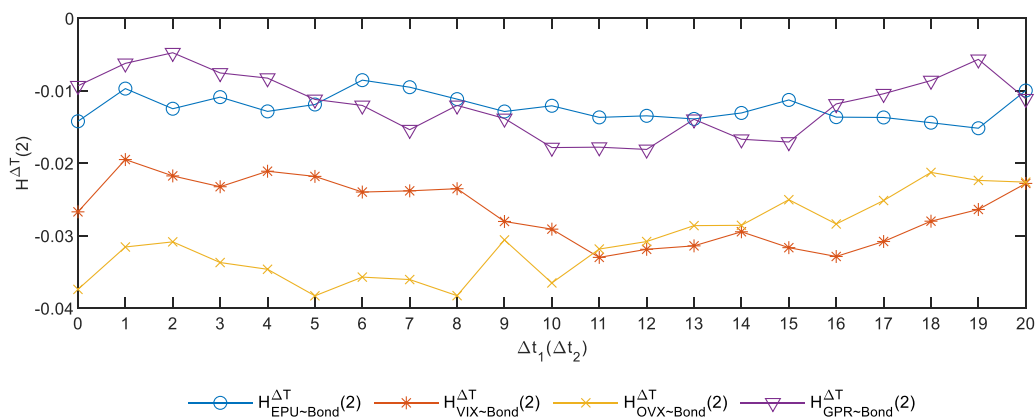


Fig. 13. Difference in the Hurst exponent at $q = 2$ between uncertainty index changes and green and conventional bond market returns with different bond market time lags.

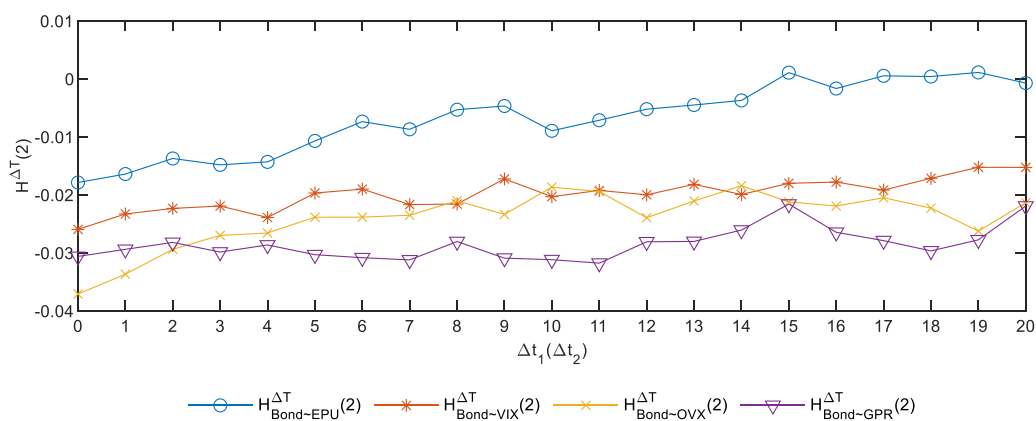


Fig. 14. Difference in the Hurst exponent at $q = 2$ between uncertainty index changes and green and conventional bond market returns with different uncertainty index time lags.

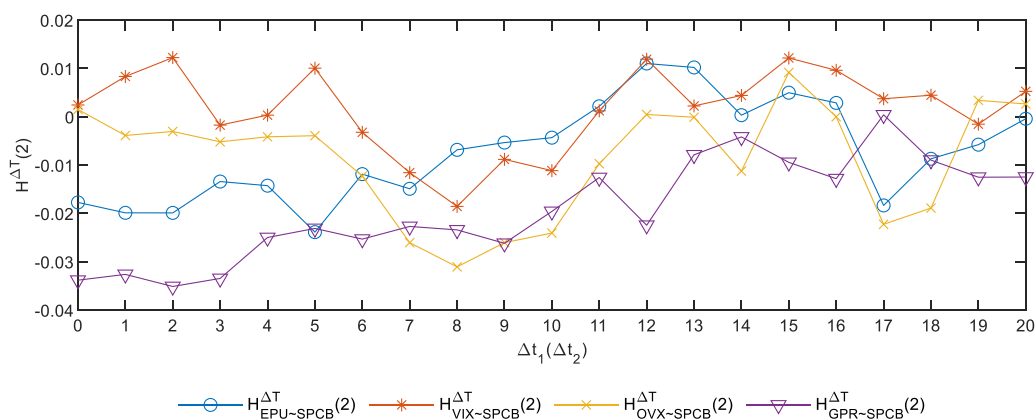


Fig. 15. Differences in two-way risk transmission between changes to uncertainty indices and conventional bond returns with different time lags.

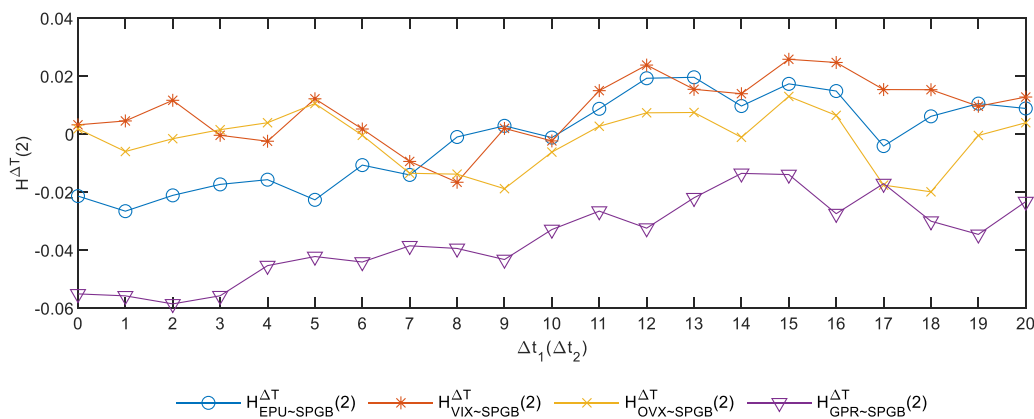


Fig. 16. Differences in two-way risk transmission between changes to uncertainty indices and green bond returns with different time lags.

have lower liquidity, which could limit its risk resilience, but this limitation is expected to improve as the market matures and develops.

5.2. Why do different uncertainty indicators have diverse relationships with the bond market?

First, different areas of uncertainty risk measure different risks. Taking geopolitical risks and EPU as examples, during the COVID-19 pandemic in 2020, the global economy faced downward pressure, resulting in a significant increase in EPU, which caused the economic uncertainty risk index to rise; however, during this period, no major geopolitical risk events resulted in a relatively low geopolitical risk index, as shown in Fig. 1. Conversely, during specific periods such as the Russia–Ukraine war, the geopolitical risk index rose significantly; therefore, the correlation between these uncertainty indicators and the bond market varies. Second, different uncertainty risks have varied impacts on the market, affecting the market on different time scales. Some uncertainty risks may have long-term effects, while others could be short-term. For instance, EPU typically signals a long-term downturn economic trend, affecting investor sentiment and overall financial market stability (Baker et al., 2016), and an economic recession would also impact the bond market. In contrast, geopolitical risks may directly impact specific commodities such as disruptions in oil supply or export risks for new energy products (Khan et al., 2024; Cheikh and Zaid, 2023). Therefore, during periods of heightened geopolitical risk, investors could opt for diversified asset allocation, transferring a portion of their assets from commodity markets to bond markets (Sohag et al., 2022). Moreover, certain geopolitical risk events have a long-term nature, which may prompt investors to prefer long-term assets such as green bonds. In contrast, a surge in crude oil and market volatility indices typically indicates higher short-term financial risks. Such volatility might spillover into the bond market, causing fluctuations (Umar et al., 2024). However, when the crude oil or stock market experiences sharp volatility, investors with lower-risk tolerance might shift a portion of their assets to relatively stable bonds as a safe haven. Since the factors causing sharp volatility in stock and crude oil markets differ, this also has varying correlations with bond markets (Wei et al., 2024).

6. Conclusion and policy implications

This study examines the correlation between four fields of uncertainty from different domains and green and conventional bond markets from a multifractal perspective to explore the issue of risk between uncertainties and bond markets, with three notable results.

First, employing the MF-ADCCA method, this study investigates the asymmetric multifractal cross-correlation between uncertainties and bond markets under different trends and investigates the characteristics between uncertainties and bond markets from the multifractal spectrum. The empirical results demonstrate varying multifractal behavior between uncertainties and bond markets (i. e., the uncertainty index has different multifractal cross-correlation with bond markets when it is in an upward or downward trend). The relationships between uncertainties in different fields and bond markets also differ, which indicates that bond markets' performance and behavior vary when facing uncertainties in different fields. The results also demonstrate that the green bond market generally has a stronger ability to avoid uncertainty risks from different fields than the conventional bond market. Large fluctuations mainly reflect the multifractal characteristics of uncertainties and bond markets. Based on these findings, investors facing uncertainty risks should focus on the different areas from which risks originate because risks from diverse areas can impact bond markets in various ways. Due to their focus on environmental projects, green bonds feature long-term gains and may exhibit different market performance compared with conventional bonds; therefore, these differences should be considered when implementing a diversified asset allocation strategy. Moreover, investors should closely monitor the rise and fall of risks from different areas because they have distinct economic implications. Certain geopolitical risks may indicate disruptions in commodity supply chains, whereas increased EPU might signal an economic downturn. Investors need appropriate control measures to adjust their asset allocation to risks originating from different areas more reasonably. Additionally, since the multifractal characteristics of uncertainties and bond markets predominantly stem from large fluctuations, special attention should be paid to sharp movements in uncertainty indices or bond markets. This will help investors

to better understand and respond to uncertainty risks, optimizing their investment decisions.

Second, using the MF-DCCA method based on risk transmission, this study further examines the risk transmission relationship between uncertainties and bond markets. The empirical results reveal differences in time-delay correlations between uncertainties in different fields and different bond markets. Specifically, when the bond market lags, EPU and geopolitical risk have an antipersistent relationship with the bond markets during the lag period, and market and crude oil volatility indices maintain a long-term persistent relationship with the bond markets during the short lag period. This indicates that bond markets must be protected against economic and geopolitical risks. Furthermore, bond markets can provide some protection from market and energy risks. When the uncertainty index lags, bond markets have a different time-delay correlation with uncertainties in different fields.

Finally, this study compares the risk transmission directions between uncertainties and green and conventional bond markets and the differing impacts of uncertainties, which further demonstrates that green bonds generally avoid risks more strongly than conventional bonds. Based on these findings, investors must be aware of the lagged effects of risks from different areas because the relationship between these risks and the bond market may change over time. Therefore, investors must be prepared to dynamically adjust their investment portfolios. The relationship between uncertainty risks from different areas and various bond markets can also differ in terms of time lags. This means that investors must be more cautious and flexible when seeking to diversify investment portfolios with bonds. Such dynamic adjustments require investors to continuously monitor risk changes and understand how such changes can impact investment strategies to make more precise and effective asset allocation decisions.

Overall, the research in this study is of great significance. Specifically, this study answers the following questions: (1) does uncertainty in different areas affect the bond market differently? and (2) do different types of bond markets behave differently under the influence of uncertainty and can green bonds serve as a diversified investment tool for conventional bonds when facing different fields of uncertainty risk? This study provides an essential reference for investors and market stakeholders to understand the impact of external shocks on bond markets and to develop targeted investment approaches and strategies based on specific areas of uncertainty. The results of this study can help policymakers and market regulators to have a clearer understanding of the risks that affect the bond market and develop strategically differentiated policies and measures to navigate different crises to ensure smooth bond market operation and development.

This paper explores differential impact of multiple risks on green and conventional bond markets from the perspective of multifractals, providing important insights for the bond markets to guard against external risks. Future research can further expand the scope of analysis by incorporating data from bond markets in more diverse economies, thereby revealing the response characteristics and differences of bond markets to uncertainty risks under different economic environments. Such an extension will not only help deepen the understanding of the operating mechanisms of the bond market but also offer more comprehensive and systematic support for cross-economy risk management and investment decision-making.

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CRedit authorship contribution statement

Shi Fengyuan: Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Guo Yaoqi:** Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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